

4(a)	protoctist ; A Protoctista I species names I eukaryote	1
4(b)	bulge in centre / swollen in centre / lost (full) biconcave shape ; ora A misshapen / shape not regular	1
4(c)	<i>any one from:</i> (merozoites) break down (haemoglobin) to produce amino acids ; (merozoites) use the amino acids for growth ; A protein used for growth (merozoites) use the amino acids to synthesise other proteins ; (merozoites) use the protein to provide energy ; I use haemoglobin for energy AVP ; e.g. red blood cells absorb water	1

Question	Answer	Marks																		
4(d)	<table><tr><th>correct sequence</th><th>step</th></tr><tr><td>1</td><td>D ;</td></tr><tr><td>2</td><td>F</td></tr><tr><td>3</td><td>B</td></tr><tr><td>4</td><td>H</td></tr><tr><td>5</td><td>A</td></tr><tr><td>6</td><td>E</td></tr><tr><td>7</td><td>G ;</td></tr><tr><td>8</td><td>C</td></tr></table>	correct sequence	step	1	D ;	2	F	3	B	4	H	5	A	6	E	7	G ;	8	C	4
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Question	Answer	Marks
4(e)	any four from: 1 <i>ref. to folding / coiling, of polypeptide chain ; in correct context of formation of secondary / tertiary structure</i> 2 contains, α- / alpha-, helix / helices or β- / beta-, pleated sheets; A β- / beta-, pleats R beta-plates CON if add incorrect idea e.g. also contains glucose 3 interactions / bonds form, between, R-groups / side chains ; R if linked to secondary structure 4 two or more named bonds (between R-groups) ; R if peptide or other bonds included <i>two from</i> hydrogen bonds disulfide, bridges / bonds I covalent bonds ionic / electrostatic, bonds hydrophobic interactions I hydrophilic interactions I van der Waals forces 5 AVP ; e.g. <i>ref. to secondary structure random coil</i> more detail of secondary structure e.g. H-bonds between (H of) N-H group and (O of) C=O group (of the polypeptide backbone) A amine / amino for NH and carboxyl for C=O (alpha-helix) pattern of H-bonds every, 3.6 amino acid (residues) / fourth amino acid / third or fourth amino acid (beta-pleated sheet) H-bonds between areas of polypeptide lying parallel more detail of tertiary structure bonds between R-groups e.g. ionic bonds between NH_3^+ and COO^- disulfide between sulfur atoms hydrophobic – between non-polar R-groups hydrogen – between oxygen atoms on C=O groups and hydrogen atoms on NH groups (of R-groups)	4

Question	Answer	Marks
4(f)	any four from: allow MSP1 for antigen 1 memory cells formed ; plus three from: 2 detail of events occurring related to clonal selection leading to memory cell formation ; e.g. phagocytosis of antigen leads to antigen presentation antigen recognition by / binding of antigen to, (specific B- / T-) lymphocytes activation / growth, of (specific) lymphocytes triggers, humoral / cell mediated, response I ref. to stimulating events occurring after clonal expansion for mp2 'MSP1 / antigen, stimulates B-cell to divide' is mp2 not mp3 3 clonal expansion / increase in numbers by many mitoses (to form memory cells) ; 4 (B- and T-) <u>memory cells</u>, long-lived / remain in circulation / AW ; I kept in body I live longer <i>unless qualified that they are present when actual antigen is present</i> 5 presence of, merozoite / actual antigen / AW, produces a <u>secondary immune response</u> ; A secondary response <i>if primary immune response introduced for mps1–3</i> <i>detail of why person, is immune / does not have symptoms of disease</i> 6 increased presence / great number, of (memory) lymphocytes increases chance of fast recognition (of antigen) AW or more / AW, of any two from, B-lymphocytes / plasma cells / T-helper cells / A B-cells T-killer cells <i>if only 'more B-cells and T-cells' stated, check for T-cells if Th and Tk types noted previously</i> 7 higher levels / faster production / AW, of antibody ; R if produced by T-lymphocytes 8 AVP ; e.g. binding of antibody to antigen prevents entry of merozoite into red blood cell T-helper cells to secrete cytokine / T-killer cells qualified <i>in context of secondary response</i>	4

Question	Answer	Marks
4(g)	any three from: 1 stop female producing eggs by preventing access to blood meals ; AW 2 example ; e.g. use (mosquito) <u>nets</u>, at night / while sleeping / to prevent blood being taken wear clothes, that cover bare skin / as a barrier / AW insect <u>repellents</u> <i>allow</i> AW if qualified e.g. on body / on clothes insect sprays, qualified e.g. prevents mosquito landing on person to feed mosquito coils / AW must be qualified as a repellent / barrier 3 example of use of, <u>insecticides / pesticides</u> / named (to kill mosquitoes) ; e.g. treat nets with insecticide / ITNs / insecticide-treated nets LLINs / long-lasting insecticide-treated (bed) nets indoor residual spraying / described A IRS use mosquito coils / insect repellents <i>allow once only</i> use of aerial sprays in, wetlands / AW 4 <i>ref. to</i> treat water to disrupt / AW, egg-laying, eggs, larvae / pupae ; 5 example ; e.g. remove bodies of water cover water tanks I cover rivers A ponds (spray) oil on, water / ponds / (small) lakes / swamps use biological control / predators / mosquitofish / other named 6 AVP ; e.g. <i>ref. to</i> use of sterile males (irradiated or genetically modified) I sterile females use of drugs to kill <i>Plasmodium</i> in (female) mosquitoes I medicine	3